

How cercit is built.

A credit appraisal system for salaried new-car loans: documents read automatically, two bureaus, versioned policy rules, a risk model, and a person who signs off.

100

database tables

460

database functions

9

AWS Lambda services

87

migrations, each tested and logged

THE PRINCIPLE

AI assists. Policy decides. A person signs off.

Four parts, kept apart on purpose, so each can be checked on its own.

1 · Document AI

What does the document say?

Reads PAN, Aadhaar, payslips, Form 16 and bank statements, and matches them against each other.

2 · Data engine

How does this person handle money?

Bureau history, income from three sources, EMIs already running, bank behaviour.

3 · Risk model

How likely is a late payment?

XGBoost, trained on synthetic loans, gives a grade and a probability.

4 · Policy engine

Does it qualify under our rules?

Deterministic, versioned, auditable. The model informs; the rules decide.

Three places, all customer data in Mumbai.

The browser

Served from GitHub Pages, holds no customer data

React 19, TanStack Router, Vite 8,
Tailwind 4

Risk model v2 in ONNX, run by
onnxruntime-web on the visitor's
device

Supabase client: every read and write
goes through checked database
functions

Supabase Postgres · Mumbai

The system of record

100 tables with row-level security
(100 policies)

Policy engine, recommendation,
bureau and income simulators, loan
book

Auth, 8 roles, 32 rights, audit log;
pg_cron jobs

AWS · Mumbai

Documents and heavy lifting

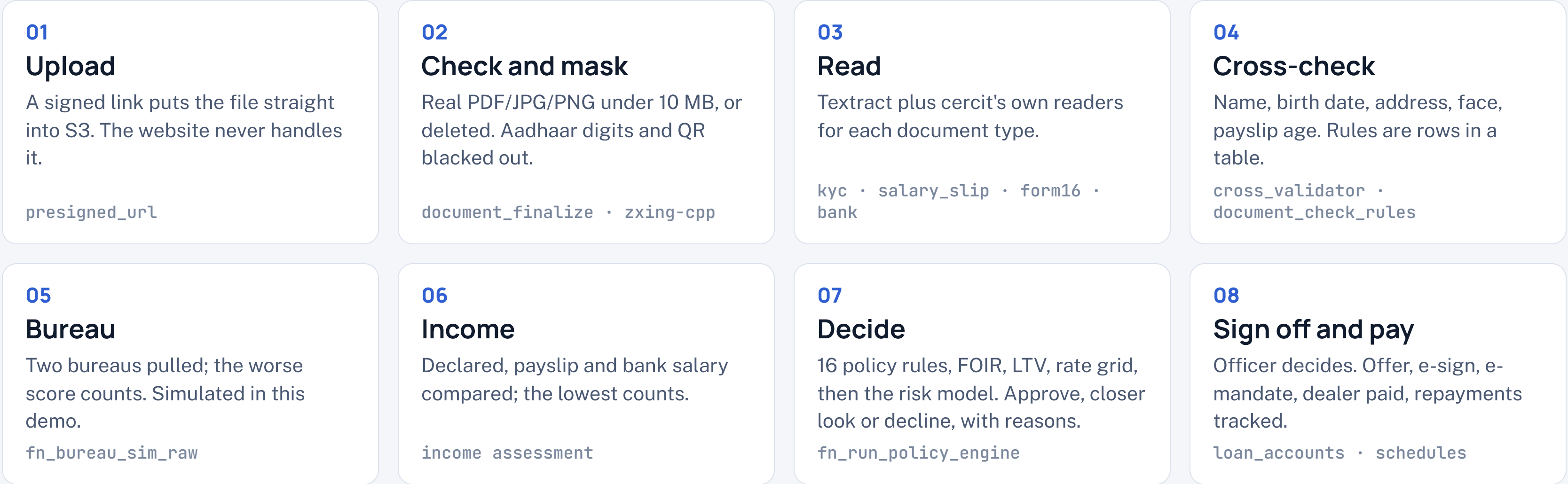
API Gateway + 9 Lambdas (Python)

S3 private bucket, signed upload links,
old versions removed

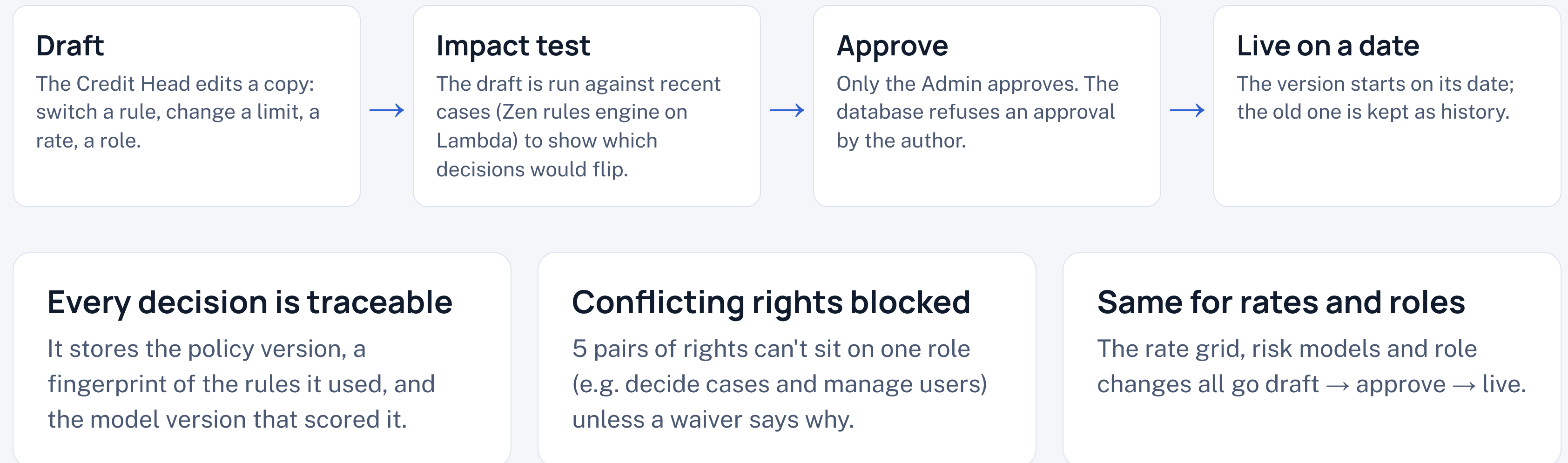
Textract reads documents ·
Rekognition matches faces

Browser → Supabase for everything about a case · Browser → AWS only to upload a file · AWS → Supabase to save what it read

From a phone upload to a decision.



Rules are data, and no one approves their own change.



RISK MODEL

A risk score that runs on the visitor's own device.

What it is

- XGBoost classifier, exported to ONNX
- Predicts the chance of going 30+ days late
- 18 inputs: bureau accounts, bank months, FOIR and more
- Version v2 is live; v1 kept for older decisions

How it was trained

- 16,995 synthetic customers, published as training data
- The live inputs are worked out the same way as the training data, and a test checks it case by case
- Gives a grade (A to E) and a probability

It informs the decision. It never overrides a policy rule.

Built as if the data were real.

Access

Least privilege

8 roles, 32 rights. Row-level security on every table. Pages read through 204 checked database functions, never straight from tables.

Personal data

Masked and encrypted

PAN and mobile encrypted at rest, shown masked. Aadhaar masked before it is stored. Full PAN only with a logged "reveal".

Location

Data stays in India

Database, documents, readers and face match all run in Mumbai (ap-south-1).

DPDP Act

Erase on request

Personal data can be erased on request, except where KYC rules require keeping it for 5 years. Every file version is deleted too.

Audit

Who did what, when

Every decision, override, rule change and sign-in is written to the audit log by the database itself.

Sign-in

Locks and timeouts

Lockout after failed tries, idle timeout per role, and two-step sign-in ready for the Admin and Head (off in this demo).

RUNNING IT

A demo that keeps moving on its own.

Every morning, 06:00 IST

A scheduled database job (pg_cron) adds 10 synthetic applications, pays out approved loans, and makes the day's repayments, including some that bounce or go late. The loan book and PAR figures stay current.

Every Sunday

A full copy of every settings, policy and pricing table is saved. No personal data is in it.

Try it as the team

Visitors sign in as Head, Manager or Officer with one click. Anything they change outside a case is journaled, and the Admin undoes it with one button.

Feedback on the way out

Signing out asks how it went, where people got lost, and what broke, with contact details only with consent.

Nothing is "done" without proof.

60

SQL test sections, run on a local Postgres (PGLite)

87

migrations in order, each logged as run and checked live

2,050

synthetic applications on the live system

635

synthetic loans in the loan book

Test first

A database change gets its test before it goes live. Lambda code has its own Python tests.

Checked live

After each change: a read-only check on the live database, a browser check, or a call to the live service.

Audited from outside

An outside AI audit was checked finding by finding; the real ones went on the fix list.

HOW IT WAS BUILT

One product person, one AI architect, a few helpers.

Who did what

- **Sameer**: product, credit policy, testing, sign-off, and every live change
- **Claude** (Claude Code): architecture, code, database, cloud, model, tests
- Lovable, Google Stitch, ChatGPT, Gemini, Copilot, Hermes, Qwen and others: prototypes, research, drafts

How the work ran

- Decide → plan in small steps → test → build → commit each step → check it live
- Live database changes are run by a person, never by the AI on its own
- No AI tool ever received real customer data

A working demo, not a licensed lender.

Simulated today

- Credit bureaus, income and bank detail, and employer lookups are simulators
- All customers and loans are synthetic
- Two-step sign-in is built but switched off for the demo

Next

- The case screen shows only the database's decision (one engine, not two)
- Missing bureau, bank or income data sends a case to a person
- Real bureau and Account Aggregator connections; commercial vehicles, three-wheelers and co-applicants

Try it at cercit.github.io/CERCIT_autoloan: pick Head, Manager or Officer on the sign-in page.